

yancc: A GPU-accelerated, differentiable solver for neoclassical transport in tokamaks and stellarators

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Abstract

We present **yancc**, a new GPU-accelerated solver for the drift kinetic equation that computes neoclassical transport fluxes, flows, and currents in tokamaks and stellarators. The drift kinetic equation is challenging to solve numerically due to strong advection-dominance, recirculating flows, internal boundary layers, severe anisotropy, and high dimensionality. Fast and accurate solutions to the drift kinetic equation are required for high fidelity optimization, profile prediction, and analysis of experiments with uncertainty quantification. **yancc** solves both the full four-dimensional drift kinetic equation (retaining speed-dependent collisions, energy scattering, and full interspecies coupling), and the reduced monoenergetic form. The discretization combines a Maxwell polynomial collocation grid in speed with finite differences in pitch angle and the flux surface coordinates, using a modified upwind stencil designed to improve diagonal dominance for multigrid efficiency. The resulting linear system is solved with a multigrid-preconditioned Krylov method. Built in JAX, **yancc** is fully differentiable, enabling gradient-based optimization and adjoint sensitivity analysis. Benchmarks against MONKES and SFINCS show agreement within 1% across a range of collisionalities, geometries, and multi-species configurations. **yancc** achieves roughly an order of magnitude speedup over SFINCS on a per-scan basis while using an order of magnitude less memory, with runtime remaining nearly flat across the full range of collisionality.

1 Introduction

Magnetic confinement fusion devices use strong magnetic fields to confine a high-temperature plasma in toroidal geometry. The magnetic field can be configured with both toroidal and poloidal components, creating nested flux surfaces along which charged particles move while radial transport across surfaces is suppressed. In stellarators, this confining field is generated primarily through external coils without requiring a toroidal plasma current, providing inherent steady-state operation and immunity to current-driven disruptions. However, the three-dimensional magnetic geometry of stellarators makes the dynamics of confined particles significantly more complex than in axisymmetric devices such as tokamaks.

Neoclassical transport theory extends classical transport theory to account for the effects of toroidal geometry. The inhomogeneous magnetic field in a torus causes particle drifts that produce populations of trapped particles that bounce between magnetic field maxima. These trapped particles, combined with guiding center drifts of the passing population, yield transport that exceeds classical predictions and depends sensitively on the collisionality regime and the structure of the

magnetic field. Solving the drift kinetic equation (DKE) gives the neoclassical particle flux, heat flux, and parallel flow for each plasma species. From these one computes the radial electric field via the ambipolarity condition, the bootstrap current, and the neoclassical conductivity, all essential inputs for transport modeling, stability analysis, and scenario development.

In stellarators, neoclassical transport differs qualitatively from tokamaks due to the three-dimensional geometry. The lack of axisymmetry creates additional ripple-trapped particle populations that can dominate transport at certain collisionalities. The radial electric field plays a crucial role: sufficiently strong $E \times B$ rotation suppresses ripple-induced transport, producing electron-root and ion-root solutions for the ambipolar electric field with transitions between them as plasma parameters vary. This ambipolar electric field must be determined self-consistently with the transport, unlike in a tokamak where ambipolarity is an intrinsic property. Accurate computation thus requires a solver that handles the full 3D geometry while resolving the sharp boundary layers that develop between different classes of trapped particles at low collisionality.

The numerical solution of the DKE is challenging due to the high resolution required for the boundary layers, the high-dimensional phase space, strong advection, anisotropic diffusion, and the need to solve the equation repeatedly for optimization or parameter scans. We therefore need a method that scales well to very high numbers of degrees of freedom, and can handle advection and anisotropic diffusion. A promising option in this direction is a multigrid approach. Multigrid methods achieve rapid convergence by operating on a hierarchy of discretizations: classical iterations efficiently damp high-frequency error but stall on low-frequency components, while multigrid overcomes this by recursively solving coarse-grid representations where low-frequency fine-grid errors appear high-frequency and are rapidly damped. Originally developed as geometric methods for elliptic problems [67, 12, 71], multigrid has since been extended to problems without a natural grid hierarchy through algebraic multigrid [57, 65]. Multigrid for advection-diffusion equations like the DKE requires additional considerations. At high Péclet numbers, strong advective coupling renders standard smoothers ineffective, motivating upwinded discretizations, line smoothers, operator-dependent transfer operators, or algebraic multigrid approaches [18, 56, 72, 54]. Multigrid has been applied to the gyrokinetic equation [1, 64], Poisson and Ampère solves in PIC simulations [14], and problems in magnetohydrodynamics [10, 2], though its application to the DKE for neoclassical transport is relatively novel.

A number of codes have been developed to solve the drift kinetic equation. DKES [36, 68, 63] and more recently MONKES [21] use a spectral Legendre expansion for the monoenergetic form, eliminating the speed coordinate and thereby dropping energy scattering and momentum conservation in exchange for performance. SFINCS [42] solves the full four-dimensional DKE (dropping only radial coupling) with the full linearized Fokker–Planck collision operator using a mixture of finite difference and spectral methods and a sparse direct preconditioner, providing high accuracy across collisionality regimes at the expense of substantial computational cost. NEO [6, 7, 5] adopts a similar approach and is primarily focused on tokamak geometry, though extensions to 3D fields also exist [8]. FORTEC-3D [62, 61] is a radially global code that uses a Monte Carlo approach to solve the DKE, and can capture full orbit width effects. PIC codes such as Euterpe [40, 41] can also be used to solve the global drift kinetic equation, though the cost is significantly higher than 4D codes such as SFINCS, making it impractical for optimization or large database studies. KNOSOS [69] uses a bounce-averaged approach to reduce the dimensionality, making it extremely fast, but contains only pitch angle scattering and is limited to low collisionality regimes, making it inaccurate when dealing with impurities. NEO-2 [38, 39] uses field line following coordinates and the full linearized collision operator, making it accurate across a wide range of collisionality, but neglects the effects of the $\mathbf{E} \times \mathbf{B}$ drift which moves particles across field lines.

Existing neoclassical solvers make unavoidable compromises between speed and physics fidelity.

The monoenergetic approximation sacrifices interspecies coupling and energy scattering, limiting accuracy for multi-species plasmas at high collisionality. Full 4D solvers like SFINCS require substantial memory and scale poorly to GPU hardware due to sparse matrix factorizations with irregular memory access. We attempt to fill this gap with **yancc** (“Yet Another NeoClassical Code”), providing a fast solver for the full DKE with minimal physics approximations by exploiting GPU parallelism for high-throughput parameter scans and optimization. Built on JAX [11], **yancc** is automatically differentiable, enabling workflows that benefit from gradients with respect to input parameters such as optimization, uncertainty quantification, and predictive transport. While a wide range of neoclassically optimized stellarator configurations have been found using existing proxy metrics [46, 73, 25, 66, 49, 30, 55, 4], recent work has benefited from including higher fidelity models in the optimization loop [52, 47, 9, 43]. Fast and differentiable transport models also benefit profile prediction workflows, and coupling **yancc** to a differentiable profile prediction code for profile optimization will be detailed in a forthcoming publication.

2 Drift kinetic equation

The drift kinetic equation [28] governs neoclassical transport in toroidal confinement devices. It has been derived a number of times in the literature [32, 34], so we only focus on the salient points here. We expand the kinetic equation for small normalized gyroradius $\rho_* = mv_{\text{th}}/(|q|B_0a)$ (here m is the particle mass, q the charge, v_{th} the thermal speed, B_0 the average field strength, and a some characteristic macroscopic length scale, taken here to be the minor radius) and take the guiding center distribution function for species s as

$$f_s = f_{1,s} + F_{M,s} \quad (1)$$

where $F_{M,s}$ is a Maxwellian, and we assume that the first order correction $f_{1,s}/F_{M,s} \sim \rho_*$ which is well satisfied in the core of tokamaks and stellarators. The guiding center distribution function depends on 5 variables, 3 in real space and 2 in velocity space. In real space we parameterize a toroidal volume by coordinates (ρ, θ, ζ) where $\rho = \sqrt{\psi_N}$ is a radial coordinate which labels flux surfaces, equal to the square root of the normalized toroidal flux ψ enclosed by the surface (not to be confused with the normalized gyroradius ρ_*), and θ and ζ are general poloidal and toroidal angles on each surface (we make no assumptions that field lines are straight in these coordinates, just that the field is tangent to surfaces of constant ρ). In these coordinates the magnetic field can be written as

$$\mathbf{B} = B_\theta \nabla \theta + B_\zeta \nabla \zeta + B_\rho \nabla \rho = B^\theta \frac{\partial \mathbf{r}}{\partial \theta} + B^\zeta \frac{\partial \mathbf{r}}{\partial \zeta} \quad (2)$$

where $\mathbf{r}$ is the position vector and the Jacobian determinant of the coordinate system is

$$\sqrt{g} = \frac{\partial \mathbf{r}}{\partial \rho} \cdot \frac{\partial \mathbf{r}}{\partial \theta} \times \frac{\partial \mathbf{r}}{\partial \zeta} \quad (3)$$

In the small ρ_* limit we take the solution to be radially local, which allows us to drop radial coupling, reducing the problem to 4D (θ , ζ , and 2 velocity space coordinates). In velocity space we choose our coordinates to be normalized speed $x_s = v/v_{\text{th},s}$ and the pitch angle $\alpha = \arccos(-v_{\parallel}/v)$. In these coordinates, the drift kinetic equation becomes:

$$\begin{aligned} \dot{x}_s \frac{\partial f_{1,s}}{\partial x_s} + \dot{\alpha} \frac{\partial f_{1,s}}{\partial \alpha} + \dot{\theta} \frac{\partial f_{1,s}}{\partial \theta} + \dot{\zeta} \frac{\partial f_{1,s}}{\partial \zeta} - \sum_{s'} C_{ss'}^\ell + S_s = -\mathbf{v}_{\text{drift},s} \cdot \nabla \rho \left(\frac{\partial F_{M,s}}{\partial \rho} \right)_{W_{0,s}} \\ + \frac{q_s}{T_s} v_{\parallel} B \frac{\langle E_{\parallel} B \rangle}{\langle B^2 \rangle} F_{M,s} \end{aligned} \quad (4)$$

Here and throughout the angle brackets $\langle h \rangle$ denote the flux surface average of a quantity h . The trajectories are given by

$$\dot{\theta} = -\frac{v_{\text{th},s} x_s \cos \alpha}{B} \frac{B^\theta}{B^2 \sqrt{g}} - \frac{B_\zeta}{B^2 \sqrt{g}} E_\rho \quad (5)$$

$$\dot{\zeta} = -\frac{v_{\text{th},s} x_s \cos \alpha}{B} \frac{B^\zeta}{B^2 \sqrt{g}} + \frac{B_\theta}{B^2 \sqrt{g}} E_\rho \quad (6)$$

$$\dot{\alpha} = -\frac{v_{\text{th},s} x_s \sin \alpha}{2B^2} \left(B^\theta \frac{\partial B}{\partial \theta} + B^\zeta \frac{\partial B}{\partial \zeta} \right) + \cos \alpha \sin \alpha \frac{1}{2B^3 \sqrt{g}} E_\rho \left(B_\zeta \frac{\partial B}{\partial \theta} - B_\theta \frac{\partial B}{\partial \zeta} \right) \quad (7)$$

$$\dot{x}_s = -(1 + \cos^2 \alpha) \frac{x_s}{2B^3 \sqrt{g}} E_\rho \left(B_\zeta \frac{\partial B}{\partial \theta} - B_\theta \frac{\partial B}{\partial \zeta} \right) \quad (8)$$

Here the radial electric field enters as $E_\rho = -\partial \Phi_0 / \partial \rho$ with Φ_0 the leading order potential that is constant on a flux surface, $v_{\text{th},s} = \sqrt{2T_s/m_s}$ is the thermal speed of species s , and B is the magnetic field magnitude. S_s is a source term that will be described below.

The collision operator $C_{ss'}^\ell$ is the Fokker–Planck–Landau operator linearized around a Maxwellian:

$$C_{ss'}^\ell = C_{ss'}[f_{1,s}, F_{M,s'}] + C_{ss'}[F_{M,s}, f_{1,s'}] \quad (9)$$

$$= C_{L,ss'} + C_{E,ss'} + C_{F,ss'} \quad (10)$$

$C_L + C_E$ make up the test particle part of the collision operator, with C_L the Lorentz pitch angle scattering operator between $f_{1,s}$ and $F_{M,s'}$:

$$C_{L,ss'} = \frac{\nu_{D,ss'}}{2 \sin \alpha} \frac{\partial}{\partial \alpha} \left[\sin \alpha \frac{\partial f_{1,s}}{\partial \alpha} \right] \quad (11)$$

and C_E the energy scattering operator between $f_{1,s}$ and $F_{M,s'}$:

$$C_{E,ss'} = \nu_{\parallel,ss'} \left[\frac{v^2}{2} \frac{\partial^2 f_{1,s}}{\partial v^2} - \frac{v^2}{v_{\text{th},s'}^2} \left(1 - \frac{m_s}{m_{s'}} \right) v \frac{\partial f_{1,s}}{\partial v} \right] + \nu_{D,ss'} v \frac{\partial f_{1,s}}{\partial v} + 4\pi \Gamma_{ss'} \frac{m_s}{m_{s'}} F_{M,s'} f_{1,s} \quad (12)$$

C_F is the field-particle scattering operator with Rosenbluth potentials $H_{s'}$ and $G_{s'}$

$$C_{F,ss'} = \Gamma_{ss'} F_{M,s} \left[\frac{2v^2}{v_{\text{th},s}^4} \frac{\partial^2 G_{s'}}{\partial v^2} - \frac{2v}{v_{\text{th},s}^2} \left(1 - \frac{m_s}{m_{s'}} \right) \frac{\partial H_{s'}}{\partial v} - \frac{2}{v_{\text{th},s}^2} H_{s'} + 4\pi \frac{m_s}{m_{s'}} f_{1,s'} \right] \quad (13)$$

$$\nabla_v^2 H_{s'} = -4\pi f_{1,s'} \quad (14)$$

$$\nabla_v^2 G_{s'} = 2H_{s'} \quad (15)$$

The collision frequencies and related coefficients are given by

$$\nu_{D,ss'} = \frac{\Gamma_{ss'} n_{s'}}{v^3} [\text{erf}(v/v_{\text{th},s'}) - \Psi(v/v_{\text{th},s'})] \quad (16)$$

$$\nu_{\parallel,ss'} = 2 \frac{\Gamma_{ss'} n_{s'}}{v^3} \Psi(v/v_{\text{th},s'}) \quad (17)$$

$$\Gamma_{ss'} = \frac{4\pi q_s^2 q_{s'}^2 \ln \Lambda_{ss'}}{(4\pi \epsilon_0)^2 m_s^2} \quad (18)$$

$$\Psi(x) = \frac{1}{2x^2} \left[\text{erf}(x) - \frac{2x}{\sqrt{\pi}} \exp(-x^2) \right] \quad (19)$$

where $\ln \Lambda_{ss'}$ is the Coulomb logarithm.

The right hand side drive term is given by

$$R_s = -\mathbf{v}_{\text{drift},s} \cdot \nabla \rho \left(\frac{\partial F_{M,s}}{\partial \rho} \right)_{W_{0,s}} + \frac{q_s}{T_s} v_{\parallel} B \frac{\langle E_{\parallel} B \rangle}{\langle B^2 \rangle} F_{M,s} \quad (20)$$

$$= -(\mathbf{v}_{\text{drift},s} \cdot \nabla \rho) \left[\frac{1}{n_s} \frac{\partial n_s}{\partial \rho} + \frac{q_s}{T_s} \frac{\partial \Phi_0}{\partial \rho} + (x_s^2 - \frac{3}{2}) \frac{1}{T_s} \frac{\partial T_s}{\partial \rho} \right] F_{M,s} \\ + \frac{q_s}{T_s} v_{\parallel} B \frac{\langle E_{\parallel} B \rangle}{\langle B^2 \rangle} F_{M,s} \quad (21)$$

$$= \left(\frac{m_s v^2}{q_s} \frac{1 + \cos^2 \alpha}{2B^3} \mathbf{B} \times \nabla \rho \cdot \nabla B \right) \left[\frac{1}{n_s} \frac{\partial n_s}{\partial \rho} + \frac{q_s}{T_s} \frac{\partial \Phi_0}{\partial \rho} + (x_s^2 - \frac{3}{2}) \frac{1}{T_s} \frac{\partial T_s}{\partial \rho} \right] F_{M,s} \\ + \frac{q_s}{T_s} v_{\parallel} B \frac{\langle E_{\parallel} B \rangle}{\langle B^2 \rangle} F_{M,s} \quad (22)$$

here the radial derivative of the leading order Maxwellian is taken at constant leading order energy:

$$W_{0,s} = \frac{m_s v^2}{2} + q_s \Phi_0 \quad (23)$$

The drive term is commonly split into three linearly independent components:

$$R_s = (A_{1,s} + x_s^2 A_{2,s})(-\mathbf{v}_{\text{drift},s} \cdot \nabla \rho) F_{M,s} + A_{3,s} B v_{\parallel} F_{M,s} \quad (24)$$

where the thermodynamic forces are given by

$$A_{1,s} = \frac{1}{n_s} \frac{\partial n_s}{\partial \rho} - \frac{q_s E_{\rho}}{T_s} - \frac{3}{2} \frac{1}{T_s} \frac{\partial T_s}{\partial \rho}, \quad (25)$$

$$A_{2,s} = \frac{1}{T_s} \frac{\partial T_s}{\partial \rho}, \quad (26)$$

$$A_{3,s} = \frac{q_s}{T_s} \frac{\langle E_{\parallel} B \rangle}{\langle B^2 \rangle}. \quad (27)$$

Given solutions to the drift kinetic equation, one can compute neoclassical fluxes and flows, the most common of which include the per-species surface average particle flux:

$$\langle \Gamma_s \rangle = \left\langle \int f_{1,s} \mathbf{v}_{\text{drift}} \cdot \nabla \rho \, d^3 \mathbf{v} \right\rangle \quad (28)$$

per-species surface average heat flux:

$$\langle Q_s \rangle = \left\langle \int \frac{m_s v^2}{2} f_{1,s} \mathbf{v}_{\text{drift}} \cdot \nabla \rho \, d^3 \mathbf{v} \right\rangle \quad (29)$$

per-species surface average parallel flow:

$$\langle V_{\parallel,s} B \rangle = \left\langle B \int v_{\parallel} f_{1,s} \, d^3 \mathbf{v} \right\rangle \quad (30)$$

the radial current:

$$\langle J_{\rho} \rangle = \sum_s q_s \langle \Gamma_s \rangle \quad (31)$$

and the parallel current (including both bootstrap and Ohmic components):

$$\langle J_{\parallel} B \rangle = \sum_s q_s \langle V_{\parallel, s} B \rangle \quad (32)$$

In addition to the full 4D problem described above, **yanc** can also solve a reduced set of equations commonly called the “monoenergetic drift kinetic equation” which is also solved by DKES [36] and MONKES [21]. For the monoenergetic form we drop all coupling in speed, reducing the problem to 3 dimensions (2 real space coordinates and pitch angle) which also makes it species independent:

$$\dot{\alpha} \frac{\partial f_j}{\partial \alpha} + \dot{\theta} \frac{\partial f_j}{\partial \theta} + \dot{\zeta} \frac{\partial f_j}{\partial \zeta} - \frac{\hat{v}}{2} \mathcal{L} f_j = s_j \quad (33)$$

The monoenergetic equation uses a modified set of trajectories to ensure conservation of the magnetic moment and to remove all direct dependence on velocity:

$$\dot{\theta} = -\frac{\cos \alpha B^{\theta}}{B} - \frac{B_{\zeta}}{\langle B^2 \rangle \sqrt{g}} \hat{E}_{\rho} \quad (34)$$

$$\dot{\zeta} = -\frac{\cos \alpha B^{\zeta}}{B} + \frac{B_{\theta}}{\langle B^2 \rangle \sqrt{g}} \hat{E}_{\rho} \quad (35)$$

$$\dot{\alpha} = -\frac{\sin \alpha}{2B^2} \left(B^{\theta} \frac{\partial B}{\partial \theta} + B^{\zeta} \frac{\partial B}{\partial \zeta} \right) \quad (36)$$

Here the monoenergetic electric field is defined as $\hat{E}_{\rho} = E_{\rho}/v$. In the collision operator we keep only the Lorentz pitch angle scattering with the monoenergetic collisionality $\hat{v} = \sum_{s'} \nu_{D, ss'}/v$. The right hand side is made up of 3 components corresponding to the three thermodynamic forces, and the equation is solved once for each right hand side:

$$s_1 = \frac{1 + \cos^2 \alpha}{2B^3} \mathbf{B} \times \nabla \rho \cdot \nabla B \quad (37)$$

$$s_2 = s_1 \quad (38)$$

$$s_3 = -\cos \alpha B \quad (39)$$

giving the three monoenergetic solutions f_1, f_2, f_3 . These are then used to compute the so-called monoenergetic transport coefficients

$$D_{ij} = \left\langle \int_0^{\pi} s_i f_j \sin \alpha \, d\alpha \right\rangle \quad (40)$$

2.1 Boundary conditions

Using generalized angle like coordinates in real space allows us to use simple periodic boundary conditions in (θ, ζ) , as opposed to more complicated conditions required when using field line following coordinates such as in KNOSOS [69]. Using the pitch angle α gives symmetric boundary conditions at $\alpha = 0$ and $\alpha = \pi$, requiring $f(-\alpha) = f(\alpha)$ and $f(\pi - \alpha) = f(\pi + \alpha)$, as opposed to a regularity type condition that appears when using $\xi = -\cos \alpha$. In speed, we require that $f \rightarrow 0$ as $x \rightarrow \infty$ which we will impose with the choice of basis for the speed coordinate in Section 3.

2.2 Properties of the drift kinetic equation

Both the forms of the drift kinetic equation have the form of a linear advection diffusion equation, for which there is a rich history of methods [51]. The prototypical equation for linear advection diffusion is

$$\mathbf{w} \cdot \nabla f - \nu \nabla^2 f = S \quad (41)$$

where $\mathbf{w}$ is the advection coefficient, commonly called the “wind” in the literature, and ν is the diffusion coefficient. Connecting the drift kinetic equation to the standard linear advection diffusion equation allows us to leverage a wide body of existing methods for such equations such as upwind finite differences and choices of multigrid smoothing and coarsening, which will be discussed further in Sections 3 and 4.

Several features of the drift kinetic equation make it more complicated than the prototypical example above. In the prototypical equation, the problem generally becomes easier for $\nu \gg 1$ as it becomes more Poisson like, while it becomes singular as $\nu \rightarrow 0$. In the drift kinetic equation the diffusive term (the collision operator) acts only in velocity space, so the diffusion is highly anisotropic – there is zero diffusion in the real space coordinates. This makes the equation singular in both limits of collisionality: as $\nu \rightarrow \infty$ we lose coupling in real space. In practice this is only a minor issue, as the normalized collisionality $\nu^* = \nu_D R_0 / v \iota$ (with R_0 the major radius, v the speed being considered and ι the rotational transform) is typically in the range $(10^{-4}, 10^{-2})$ though for impurities it can be several orders of magnitude larger.

Another important feature of the drift kinetic equation is that all collisionless characteristics at $E_\rho = 0$ are effectively closed. This is strictly true for trapped particles, as well as passing particles on rational surfaces. On irrational surfaces, the trajectories approximately close after a sufficient number of transits. The periodic boundary conditions in real space combined with the symmetry/reflecting boundaries in pitch angle means that no information enters or leaves the domain, which will affect the choice of smoothing operation in the multigrid scheme, described in Section 4.2. At finite E_ρ , the orbits generally don’t close exactly, but are still bounded and recurrent, so similar smoothing properties apply.

One can show [42] that the drift kinetic equation has a null space of dimension $2N_s$ where N_s is the number of species consisting of $f_{1,s} = c_s F_{M,s}$ and $f_{1,s} = c_s v^2 F_{M,s}$ where c_s is an arbitrary constant. The monoenergetic form has a null space of dimension 1, corresponding to $f_j = c$ for some constant c . To remove this null space, we impose additional gauge constraints. For the full drift kinetic equation, we require that all of the surface averaged density and pressure reside in the leading order Maxwellian for each species:

$$\left\langle \int f_{1,s} d^3 \mathbf{v} \right\rangle = 0 \quad (42)$$

$$\left\langle \int v^2 f_{1,s} d^3 \mathbf{v} \right\rangle = 0 \quad (43)$$

In addition to this gauge freedom, one can also show (see Sec. III of [42]) that the steady state drift kinetic equation requires an additional source for it to be solvable when E_ρ is nonzero and not at its ambipolar value (i.e., E_ρ such that $J_\rho = 0$). This is due to nonzero E_ρ being a singular perturbation to the $E_\rho = 0$ equation. To account for this we follow [42] and add additional terms that serve as a source of particles and heat for each species to ensure solvability for arbitrary E_ρ . For the trajectory models considered here the particle source is not strictly necessary (the source is always zero at the solution) but it helps by making the overall system square, and is needed

when alternative trajectory models are considered [42]. Both source terms vanish when E_ρ is at its ambipolar value, so they should be viewed merely as a numerical expedient, and do not affect the true transport physics. The source term for each species has the form:

$$S_s(x_s) = S_{s,p} F_{M,s}(x_s)(x_s^2 - \frac{5}{2}) + S_{s,h} F_{M,s}(x_s)(x_s^2 - \frac{3}{2}) \quad (44)$$

where $S_{s,p}$ and $S_{s,h}$ are the free parameters to be solved for.

For the monoenergetic equation, the only gauge freedom that exists is the choice of the average of f_j . One could impose an additional constraint requiring, e.g., zero average, but it is simpler to just require that f take a fixed value at a given point:

$$f_j(\alpha = \pi/2, \theta = 0, \zeta = 0) = \text{const} \quad (45)$$

The final important feature, which also appears in the prototypical equation, is the presence of internal boundary layers that develop in the solution with a characteristic width $\sqrt{\nu}$. These can be seen clearly in Figure 3 of [3] or Figure 5 of [44], and correspond to the accumulation of particles near transitions between trapping regions [35, 37], necessitating very high resolution in the real space and pitch angle coordinates.

3 Discretization

Because the location of boundary layers doesn't depend on speed x_s , we can often get away with a relatively coarse discretization in the speed coordinate. Following SFINCS [42], we represent $f_{1,s}$ in speed using weighted Maxwell polynomials (which we denote L_k), which are orthogonal on $(0, \infty)$ with respect to the weight function e^{-x^2} [45]. In practice we usually only need 5–10 points in the speed coordinate with this discretization. The advective term $\dot{x}_s \partial f_{1,s} / \partial x_s$, the pitch angle scattering operator C_L , and the field-particle operator C_F are collocated at the Maxwell-polynomial nodes using the associated pseudospectral differentiation matrices.

The energy scattering operator C_E is instead discretized in weak (Galerkin) form. Collocating C_E pointwise can lead to spurious anti-dissipative modes. The lowest speed node has no neighbor below it, so the one-sided stencil for its second derivative is anti-diffusive, and is stabilized only by a drag term that is normally positive at that node. When a background species has a thermal speed that places its collisional transition near the lowest speed node of the kinetic species, this drag term can change sign, leaving a growing mode whose magnitude is comparable to the stiffest resolved mode. Because the multigrid smoothers used in Section 4.2 require the discrete operator to be dissipative, this defect stalls convergence for otherwise unremarkable combinations of species masses, temperature ratios, and collisionality.

We avoid this by writing $C_{E,ss'}$ in flux form,

$$C_{E,ss'} f_{1,s} = \frac{1}{x_s^2} \frac{\partial}{\partial x_s} \left[a_{ss'}(x_s) \left(\frac{\partial f_{1,s}}{\partial x_s} + 2\tau_{ss'} x_s f_{1,s} \right) \right], \quad a_{ss'} = \frac{\nu_{\parallel,ss'} x_s^4}{2}, \quad \tau_{ss'} = \frac{T_s}{T_{s'}}, \quad (46)$$

which is algebraically identical to the operator given in Section 2. Testing this against each Maxwell polynomial and integrating by parts gives a discrete operator built from a mass matrix and a stiffness matrix,

$$\int_0^\infty L_k x_s^2 \frac{\partial f_{1,s}}{\partial t} e^{-x_s^2} dx_s = - \int_0^\infty \frac{\partial L_k}{\partial x_s} a_{ss'} \left(\frac{\partial f_{1,s}}{\partial x_s} + 2\tau_{ss'} x_s f_{1,s} \right) e^{-x_s^2} dx_s, \quad (47)$$

with the boundary term vanishing because $a_{ss'} \sim x_s^4$ as $x_s \rightarrow 0$, and both integrals evaluated with the same Gaussian quadrature used for the collocation nodes. For $\tau_{ss'} = 1$ this gives a positive-definite mass matrix and a symmetric positive-semidefinite stiffness matrix, so the discrete operator is dissipative for any combination of colliding species; empirically we find this remains true for $\tau_{ss'} \neq 1$ as well.

One could consider using a fully spectral method for the drift kinetic equation, which has been successfully used for the monoenergetic form in DKES and MONKES, however we found that these did not scale well when applied to the full 4D problem. Due to the sharp boundary layers that develop at low collisionality, any method requires a large number of grid points or spectral collocation points to sufficiently resolve these, so the usual reduction in resolution possible with spectral methods for smooth solutions isn't possible. Additionally, spectral methods tend to give dense matrices (though using Legendre polynomials for pitch angle results in a block pentadiagonal system, the block size would be $N_s n_x n_\theta n_\zeta$ which can be quite large).

Because of this, in all other coordinates we adopt a finite difference discretization with a defect correction approach where we use a 2nd order scheme for the preconditioner and a 4th order scheme for the main operator. We use uniform grids in α, θ, ζ , with full index grids in θ, ζ and half index grid in α :

$$\alpha = \frac{\pi(2j+1)}{2n_\alpha} \quad j \in \{0, 1, \dots, n_\alpha - 1\} \quad (48)$$

$$\theta_j = \frac{2\pi j}{n_\theta} \quad j \in \{0, 1, \dots, n_\theta - 1\} \quad (49)$$

$$\zeta_j = \frac{2\pi j}{N_{FP} n_\zeta} \quad j \in \{0, 1, \dots, n_\zeta - 1\} \quad (50)$$

where N_{FP} is the number of field periods of the underlying equilibrium (a discrete toroidal symmetry common in stellarators). We use standard centered differences for the collision operator, with the Lorentz pitch angle scattering operator being explicitly split into first and second derivative parts, with centered differences applied to each:

$$\mathcal{L} = \frac{1}{\sin \alpha} \frac{\partial}{\partial \alpha} \sin \alpha \frac{\partial}{\partial \alpha} = \frac{\partial^2}{\partial \alpha^2} + \frac{\cos \alpha}{\sin \alpha} \frac{\partial}{\partial \alpha} \quad (51)$$

Using the half index grid in α avoids the singular behavior at $\alpha = 0$ and $\alpha = \pi$. To enforce boundary conditions in α we mirror f around the endpoints which explicitly enforces $f(-\alpha) = f(\alpha)$ and $f(\pi - \alpha) = f(\pi + \alpha)$.

For the advective terms we use an upwinded stencil. The “standard” 2nd/4th order upwind scheme uses stencil points $(0, 1, 2)$ and $(0, 1, 2, 3, 4)$ respectively, where these integers give the relative grid indices for a forward difference, with 0 being the point the derivative is being evaluated at, positive numbers are grid points to the right, negative to the left; indices are reversed about 0 for backward differences. In contrast, we use a modified stencil containing the points $(0, 1, 4)$ (2nd order) and $(-2, 0, 1, 3, 4)$ (4th order). These wider stencils give a larger element on the diagonal relative to the off-diagonals, resulting in matrices with a higher degree of diagonal dominance. We quantify this diagonal dominance with the ratio d :

$$d = \frac{|a_{ii}|}{\sum_{j \neq i} |a_{ij}|}. \quad (52)$$

A matrix with $d > 1$ is diagonally dominant in the traditional sense. In Table 1 we compare the stencils and coefficients for our widened 2nd and 4th order schemes vs the standard upwinded

stencils, comparing the degree of diagonal dominance along with the magnitude of the leading order error constant. We find that these more diagonally dominant stencils significantly improve the multigrid smoothing (see Section 4.2) with negligible loss in accuracy.

Scheme	Stencil	Coefficients	d	leading error
standard 2nd order	(0, 1, 2)	$\frac{1}{h}(-\frac{3}{2}, 2, -\frac{1}{2})$	0.60	0.33
wide 2nd order	(0, 1, 4)	$\frac{1}{h}(-\frac{5}{4}, \frac{4}{3}, -\frac{1}{12})$	0.88	0.67
standard 4th order	(0, 1, 2, 3, 4)	$\frac{1}{h}(-\frac{25}{12}, 4, -3, \frac{4}{3}, -\frac{1}{4})$	0.24	0.20
wide 4th order	(-2, 0, 1, 3, 4)	$\frac{1}{h}(-\frac{1}{15}, -\frac{13}{12}, \frac{4}{3}, -\frac{4}{15}, \frac{1}{12})$	0.62	0.20

Table 1: Comparison of the widened upwind stencils used in `yancc` against the standard upwind stencils, showing the diagonal dominance ratio d and the magnitude of the leading order error constant. Our widened stencils have a larger relative magnitude on the diagonal for similar error.

An important requirement when considering discretizations for a multigrid scheme is to ensure numerical stability not just at the finest grid level but at all levels. Using upwind finite differences achieves this by adding artificial numerical diffusion to avoid small grid scale oscillations that can happen when under-resolving sharp features. This increased numerical stability is not without cost though. Because the additional diffusion is proportional to the grid spacing, coarser grids effectively have higher diffusion, which can make them a poor approximation to the true fine grid problem, causing convergence issues for a multigrid scheme. This is a well known problem when using multigrid for advection diffusion problems [67] and techniques to mitigate it will be discussed in Section 4.3.

The integrals required for output moments are computed with spectral accuracy in all coordinates. In x we use the Gaussian quadrature weights associated with the Maxwell polynomials, while in θ, ζ we use standard trapezoidal quadrature which is spectrally accurate for periodic integrands. In α we use Fejér type 1 quadrature which is equivalent to Chebyshev quadrature in $\xi = -\cos \alpha$.

For the field part of the collision operator, there are a number of possible ways to discretize the Rosenbluth potentials. One could treat them as additional unknowns and solve a system of equations for f, G, H , but we adopt a Green’s function approach to analytically solve the Poisson equations for G, H [50]. The details of this derivation are given in Appendix A.

4 Multigrid method and Krylov solver

When discretized, the drift kinetic equation becomes a large, sparse linear system of the form $Af = s$. The matrix A is banded in θ and ζ (due to the finite difference discretization), and dense in α and x (due to the spectral collocation in x and the integral terms in the collision operator for α). The monoenergetic equation is banded in all three coordinates (α, θ, ζ) . Due to the high resolution required to resolve the trapped passing boundary, the resulting system typically has $\sim 10^6$ – 10^7 degrees of freedom ($\sim 10^5$ is typical for the monoenergetic problem). At this size for this problem, direct sparse solvers were found to be quite poor, with `scipy`’s `spsolve` [70, 19] failing to solve even the monoenergetic problem in a reasonable time – for a problem with 137k unknowns (still an order of magnitude less than what is commonly required for the DKE), `spsolve` needed more than 30 minutes, while on the same hardware, using the methods outlined below, `yancc` needed 18 seconds.

A more efficient approach is to use a Krylov method like GMRES combined with a good preconditioner. Existing codes like SFINCS use a sparse LU factorization of an approximate A as a preconditioner, which tends to work well at low to moderate collisionality, at the cost of a signif-

icant amount of memory (often 100s of GB) to perform the sparse factorization. Additionally, by dropping certain terms from the preconditioner, it becomes far less effective at high collisionality, requiring many more Krylov iterations. Finally, sparse matrix factorizations tend to see minimal speedup on GPU, due to the irregular memory access [60, 22, 15].

We opt for a different approach, using a multigrid preconditioner combined with an outer Krylov solver. The main elements of a multigrid scheme are a hierarchy of grids, a smoothing operation, and a coarse grid correction. The basic idea is as follows:

1. Starting from some initial guess on a fine grid, apply a smoothing operation which reduces the high-frequency components of the error.
2. Restrict the fine grid problem down to a coarser grid. Because the smoothing eliminated the high-frequency error, the coarse grid problem now captures the dominant error.
3. Solve the coarse grid problem (possibly approximately) to obtain a correction.
4. Interpolate this correction back to the fine grid.
5. Repeat.

The key idea is that step 3 can be done recursively, using additional coarser levels to reduce the problem size until it is small enough to solve directly. In the ideal case, multigrid methods can achieve a grid-independent rate of convergence, where each pass through the multigrid cycle reduces the error by a fixed fraction, regardless of resolution, meaning that the number of iterations required to achieve a given error tolerance is constant as the resolution is increased (compared to many other methods which require a larger number of iterations at higher resolution). For advection-diffusion problems obtaining ideal multigrid convergence is generally difficult, but even in this case it can be a highly efficient preconditioner.

4.1 Grid hierarchy

The first ingredient in a multigrid scheme is a hierarchy of different grids and a method to transfer information between them. The resolution on the finest grid is chosen to properly represent the physics of the problem with our given discretization. We then must define how the coarse grids are constructed. Because even the finest grid generally requires very low resolution in x ($n_x \sim 5\text{--}10$) we find little benefit in coarsening the x coordinate, so we retain the finest x grid across all levels. This is a form of “semi-coarsening” where one only coarsens the grid in certain directions. In the other coordinates (α, θ, ζ) we coarsen by a factor of 2 in each direction each time we go down a grid level, for a total reduction in cost of roughly 8 per grid level. We find that increasing the coarsening factor in each coordinate to 3 or even 4 only moderately degrades solver performance. Semi-coarsening in other coordinates was also considered but was found to be significantly more expensive in time and memory per iteration (due to the correspondingly larger coarse grids) without a significant decrease in the number of iterations.

The final choice in grid hierarchy is when to stop coarsening. For simple problems one can coarsen extremely far, until the coarsest grid contains only a single node. In our case we find that using a larger coarse grid improves performance. Typically we coarsen until the coarsest grid has a few thousand degrees of freedom at which point we directly solve the coarsest grid system with a dense LU factorization. Going coarser than this is generally inefficient, since dense linear algebra on GPU is very fast and each additional grid level adds overhead. Using an exact solve on a larger grid also reduces the effects of the additional numerical diffusion on the coarse grid discussed in Section 4.3.

In addition to the grid hierarchy, we must define operations for moving information between grids. These are generally referred to as restriction (moving from fine to coarse) and prolongation (from coarse to fine). There are a number of methods for restriction/prolongation in use in the literature. A common approach is to use operator dependent methods [18], where one considers the direction of flow and diffusive coupling to determine how to move between grids. We find in our case that simple piecewise linear interpolation is sufficient for prolongation, with higher order interpolation or operator dependent methods increasing the per iteration cost and complexity without a noticeable decrease in the required number of iterations. For restriction we use the volume weighted transpose of the prolongation operator.

4.2 Smoothing

The next key ingredient for a multigrid scheme is an efficient smoothing operation. For linear problems, this typically takes the form of a classical relaxation method which takes the form

$$f_{k+1} = f_k + \omega_s K^{-1}(s - Af_k) \quad (53)$$

where f is the unknown being solved for, s is the right hand side, k is an iteration index, K is some approximation of A and ω_s is a damping/under-relaxation parameter. For Poisson type problems the standard choice is $K = \text{diag}(A)$, however for advective or anisotropic diffusion problems this is generally poor due to strong coupling along characteristics which is ignored by the diagonal approximation. For advection problems, it is common to take K as the upper or lower triangle of A , or a block triangular approximation where one keeps full coupling in some coordinates and triangular coupling in others. For problems with open characteristics and Dirichlet boundary conditions this can be extremely efficient, as the smoothing operation takes the known boundary value and sweeps it in along characteristics, reducing the error at all frequencies across the entire domain. As discussed previously however, the drift kinetic equation has periodic boundary conditions and a wide variety of closed characteristics. Because of this, instead of a triangular approximation of A we instead use a block diagonal approximation for smoothing. This is significantly cheaper to apply (especially on GPU) and performs just as well in practice.

When using a block diagonal approximation, the choice of coordinate ordering determines which matrix elements are on the block diagonal and are thus retained in the smoothing operation. To avoid having to make an explicit choice, we consider a family of block diagonal smoothers, which each maintain full coupling along one coordinate [56, 20]. If A is our original matrix, and P_j is a permutation matrix that reorders the coordinates to put coordinate $j \in \{x, \alpha, \theta, \zeta\}$ on the block diagonal, then we take $K_j = P_j^{-1} \text{block_diag}(P_j A P_j^{-1}) P_j$, giving 4 smoothers (3 for the monoenergetic problem) which each preferentially smooth the errors in a different direction. One full smoother pass consists of applying these 4 smoothers in sequence to a given estimate of the solution.

This choice of block diagonal smoothing matrices is aided by our choice of finite difference stencil (Table 1), which makes the matrices more diagonally dominant compared to standard upwinded methods. This larger element on the diagonal means a block diagonal approximation captures more of the behavior of A , resulting in a more efficient smoother. We can measure the efficiency of the smoother by considering the operator that multiplies the error at each smoothing step:

$$f_{k+1} - f_{\text{exact}} \equiv e_{k+1} = (I - \omega_s K^{-1} A) e_k \quad (54)$$

We can then consider

$$r = \frac{\|L(I - \omega_s K^{-1} A)\|}{\|L\|} \quad (55)$$

where L is some approximate high pass filter, which following [26] we take to be the discrete Laplacian on our phase space. This ratio r gives a measure of how much the smoother is reducing the high-frequency errors (lower is better). In Figure 1 we compare this ratio for the block diagonal smoothers in each coordinate for our upwinded stencil vs the “QUICK” method (Quadratic Upstream Interpolation for Convective Kinematics, from [48]) on the monoenergetic problem. In both cases we first optimize the damping parameter ω_s to achieve the lowest r possible for each collisionality considered across a range of different geometries and electric fields. We see that our method allows a larger damping parameter and gives much lower smoothing factor, indicating that more of the high-frequency error is eliminated per application, especially at low collisionality for θ , ζ and across all collisionalities for α . For the full drift kinetic equation, which spans several orders of magnitude in collisionality, we find that a fixed damping parameter of $\omega_s = 0.6$ works best.

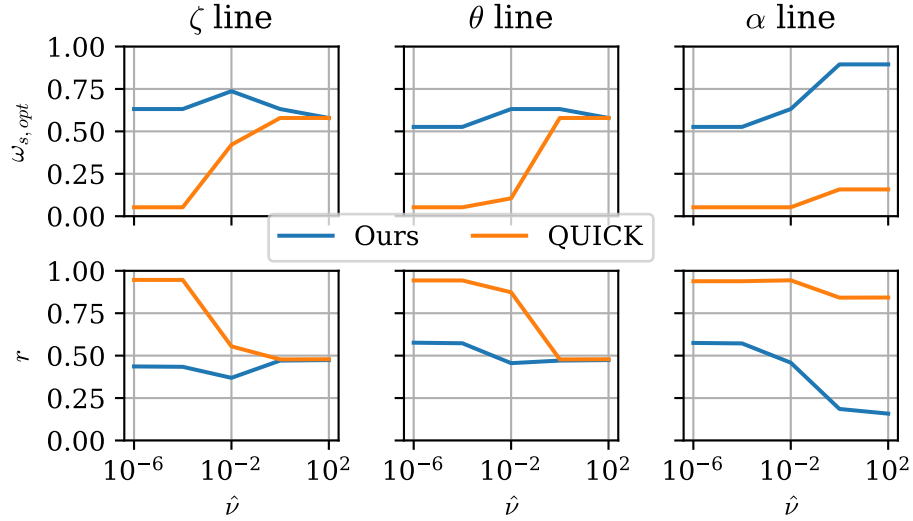


Figure 1: Smoothing factor r for the block diagonal smoothers in each coordinate, comparing the widened upwind stencil used in `yanc` against the QUICK scheme on the monoenergetic problem, as a function of collisionality. Using a wider stencil with a larger diagonal significantly improves the smoothing property.

Because we use a finite difference discretization, the block diagonal itself is banded for permutations that couple θ and ζ , and approximately banded for permutations that couple α (for the smoothing operation we drop the off band terms corresponding to the integrals in the field part of the collision operator). This significantly reduces the memory required to store the smoothers, and due to the larger elements on the diagonal we find that an unpivoted banded LU factorization is stable, which avoids fill in and irregular memory access.

4.3 Coarse grid problem

The standard coarse grid correction takes the form of

$$f_{k+1,\text{fine}} = f_{k,\text{fine}} + \omega_c P A_{\text{coarse}}^{-1} R(s_{\text{fine}} - A_{\text{fine}} f_{k,\text{fine}}) \quad (56)$$

where R is the restriction operator from fine grid to coarse grid, P is the prolongation operator from coarse grid to fine grid, and as in the smoothing we allow for a damping parameter ω_c .

An important consideration in a multigrid scheme is the choice of how to discretize the coarse grid problem. The two primary methods are direct discretization, and Galerkin projection. In the direct method, we simply re-discretize the original problem on the coarser grid, using the same underlying finite difference method. This has the advantage of being cheap to implement and apply. Galerkin projection takes

$$A_{\text{coarse}} = RA_{\text{fine}}P \quad (57)$$

This is commonly used in algebraic multigrid schemes and is often recommended when dealing with advective problems, however it can require more memory and be difficult to implement in a matrix free manner. The additional memory is due to the Galerkin projection changing the sparsity structure, meaning that in many cases A_{coarse} can have more nonzero elements than A_{fine} , especially when combined with semi-coarsening. This also makes coarse grid operations proportionally more expensive. Additionally, in our case we wish to avoid ever materializing even a sparse representation of A_{fine} to keep memory usage low. This is easy when doing direct re-discretization, as applying A_{fine} and A_{coarse} consist of convolutions, small tensor contractions and elementwise operations, with A_{coarse} being orders of magnitude smaller and thus cheaper. With a Galerkin projection however, applying A_{coarse} in a matrix free manner requires interpolating the coarse grid residual back to the fine grid, applying A_{fine} , then restricting. This means that the cost of the coarse grid operations is as expensive as the fine grid, which significantly increases the cost of the multigrid cycle. For these reasons we opt for direct rediscretization which we find works sufficiently well in practice.

4.4 Cycle type

The final element of the multigrid scheme to consider is the cycle type, governed by the “cycle index” which we denote γ . A cycle index of $\gamma = 1$ corresponds to a “V” cycle, where we visit each coarser grid once per cycle. Cycle index $\gamma = 2$ corresponds to a “W” cycle, where we visit each coarser grid twice, etc. These are shown diagrammatically in Figure 2. We find that for the drift kinetic equation $\gamma = 1$ is the most robust. As discussed above, the coarse grid correction can be unreliable due to additional numerical diffusion, so higher order cycles that apply this less accurate coarse grid correction more often tend to slow the solver down. For the monoenergetic equation however, we find that this is less of an issue and using $\gamma = 3$ tends to be most efficient.

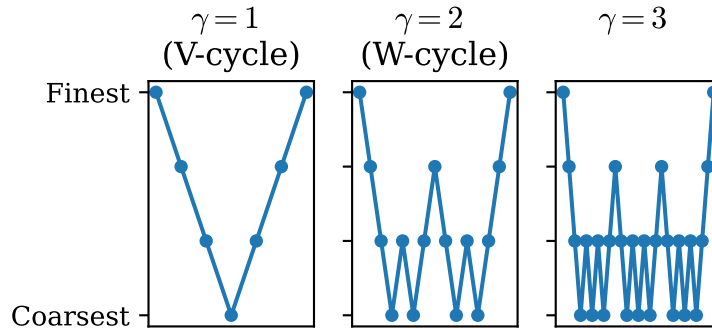


Figure 2: Schematic of multigrid cycle types for different cycle index γ (V-, W-, and higher cycle indices) across the grid hierarchy.

4.5 Krylov solver

For many problems a well tuned multigrid method works efficiently as a standalone solver, however this is often not the case for advection diffusion equations. This can be seen in Figure 3, where we plot the spectrum of the discretized drift kinetic equation A for a representative example, as well as $I - MA$ where M is the multigrid preconditioner. This matrix is what multiplies the error at each step of a pure multigrid cycle. If all of its eigenvalues are within the unit circle then pure multigrid would converge. We see that in practice, the multigrid preconditioner clusters the vast majority of the eigenvalues near zero, which means they are very efficiently damped. However there are several eigenvalues near the unit circle which are only weakly damped which would slow convergence, and there can also be outliers that would cause divergence in a pure multigrid scheme. To handle this, we use multigrid as a preconditioner for a Krylov solver. We use a right-preconditioned GCROT [17, 33] with inner GMRES [59, 58]. The largest memory requirement is typically storing the Krylov subspace, so we limit the inner GMRES to 150 iterations before restarting. This is more than enough for the majority of cases tested, and the outer GCROT loop efficiently handles restarting by preserving the most important elements of the Krylov subspace between restarts rather than starting from scratch as in standard restarted GMRES.

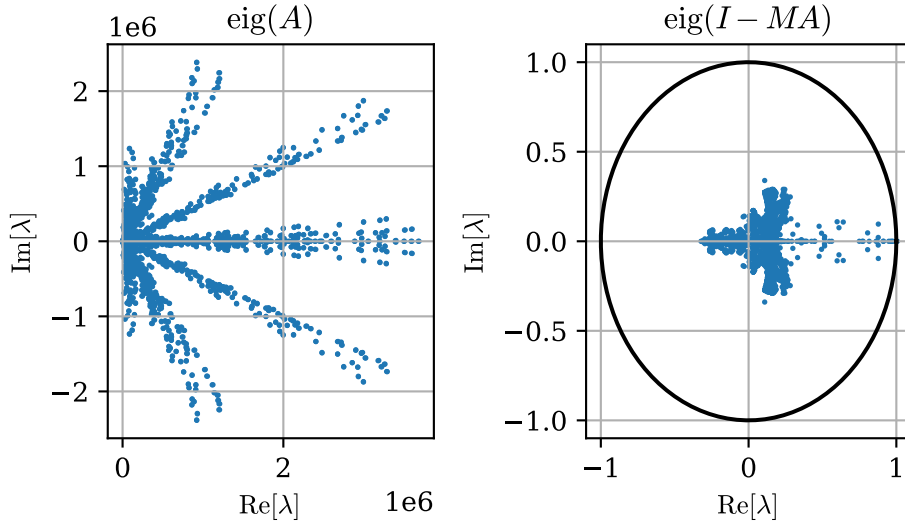


Figure 3: Spectrum of the discretized drift kinetic operator A for a representative example, and of the preconditioned error operator $I - MA$, where M is the multigrid preconditioner. λ denotes the eigenvalues of the discretized operator. Choosing different parameters for the DKE (such as a different magnetic field) will give slightly different but qualitatively similar spectra.

4.6 Ambipolar rootfinding

In a stellarator, the particle transport is not intrinsically ambipolar like in a tokamak, meaning that the radial current $J_\rho = \sum_s q_s \Gamma_s$ is not zero for arbitrary electric field E_ρ . This means that to obtain physically meaningful solutions, one must find the ambipolar value(s) of E_ρ self consistently. To do this, we use Newton's method to find the roots of $J_\rho(E_\rho) = 0$. Each Newton step then requires solving the drift kinetic equation for a fixed value of E_ρ and its derivative, which is supplied by automatic differentiation. In many cases only a single root exists, but in exceptional cases there can

be three roots [27], corresponding to so-called electron and ion branches, with electron roots being favorable for confinement [31]. To find multiple roots, we apply deflation, where after each root E_ρ^i is found, we divide it out and find the roots to the modified equation:

$$J_\rho(E_\rho) \prod_i \frac{1}{E_\rho - E_\rho^i} \quad (58)$$

After finding a root of the deflated system, we use that as an initial guess for the original system to ensure it is a true root and not a numerical artifact of the deflated equation. To improve robustness we also make use of bracketing, where we keep track of sign changes during iterations, and use these to bracket roots and prevent the Newton steps from diverging when we know a root exists within a given domain, falling back to bisection if the Newton step fails to make sufficient progress.

5 Benchmarks

5.1 High aspect ratio tokamak

As a first example we compare **yanc** to analytic results for a high aspect ratio circular tokamak. The parameters are given in Table 2. Using a single ion species means there is no parallel friction, and several transport coefficients then take values that are exact at any collisionality and aspect ratio. Because like-particle collisions conserve momentum, a single species cannot transport itself radially, so the radial particle flux vanishes for every drive. The density and radial-electric-field drives enter the kinetic equation only through the combination $A_1 = \partial_\rho \ln n + (Ze/T)\partial_\rho \Phi - \frac{3}{2}\partial_\rho \ln T$, which is isotropic in energy, so they drive no conductive heat flux and produce exactly the diamagnetic flow $\langle BV_\parallel \rangle = -A_1 KT/(Ze)$ with $K = G/(\iota \partial_\rho \psi)$. They are also degenerate with one another, the $\mathbf{E} \times \mathbf{B}$ drift being negligible at this field strength. Each drive is solved separately, the electric field one at the small E_ρ of Table 2 and rescaled by $T/(ZeE_\rho)$, so that its column is the response per unit A_1 and shares the normalization of the density column.

ρ	0.5
$B(\theta)$	$B_0/(1 + \epsilon \cos \theta)$
B_0	1 T
R_0	3 m
$q = 1/\iota$	2
G, I	$B_0 R_0, 0$
$\epsilon = \rho a/R_0$	0.01
T	1 keV
n	$\in [2.0 \times 10^{15}, 6.3 \times 10^{22}] \text{ m}^{-3}$
$\ln \Lambda$	17
E_ρ	0.1 V, rescaled to unit A_1
$\nu_{i,\text{Sauter}}^* = \frac{qR_0}{\epsilon^{3/2} v_T \tau_i}$	$\in [10^{-3}, 3.2 \times 10^4]$
Species	Deuterium
yanc resolution ($n_x, n_\alpha, n_\theta, n_\zeta$)	(6–16, 121–3201, 61–121, 1)

Table 2: Parameters and resolutions for the circular tokamak comparison with analytic theory. The configuration is axisymmetric, so a single toroidal grid point is used. The density sets the collisionality at fixed temperature, and the resolution varies with collisionality as described below.

Figure 4 shows the resulting transport matrix across eight decades in collisionality, spanning both asymptotic regimes. The five quantities that are exact are reproduced to 10^{-3} or better in the normalizations shown. At the collisional end the heat flux approaches the Pfirsch–Schlüter value [32] to 0.2% and the flow approaches Hazeltine’s coefficient [29] to 1.3%, the latter well inside that formula’s own accuracy of roughly 20%. At the collisionless end the flow reaches the banana-regime coefficient $1.17f_c$ (with f_c the circulating particle fraction) to within 0.5%, while the heat flux exceeds the $\epsilon \rightarrow 0$ value by 15%; a separate scan in aspect ratio confirms that this is the expected $O(\sqrt{\epsilon})$ finite-aspect-ratio correction, the measured coefficient falling from 2.08 at $\epsilon = 0.3$ to 0.99 at $\epsilon = 10^{-3}$ and extrapolating linearly in $\sqrt{\epsilon}$ to the analytic 0.92. The two regimes make opposite demands on the discretization. In the banana regime the trapped-passing boundary layer sets $n_\alpha \gtrsim 16/\sqrt{2\epsilon\nu_i^*}$, while at high collisionality the requirement is one of momentum conservation: pitch-angle scattering is discretized by finite differences and so conserves parallel momentum only to $O(n_\alpha^{-2})$, an error amplified by ν_i^* because the parallel viscosity it competes against is itself small, $O(\epsilon^2/\nu)$. Recovering the exact flow there requires $n_x \geq 16$ and n_α growing as $\sqrt{\nu_i^*}$.

5.2 W7-X monoenergetic DKE

For the next example we solve the monoenergetic drift kinetic equation for the high mirror configuration [24] of W7-X over a range of collisionality and electric field. The results are shown in Figure 5, comparing **yancc** vs MONKES [21]. Both codes are converged to within 1% across the full range of collisionality. The resolutions for each code are given in Table 3.

ρ	0.45
$\hat{\nu} = \nu_D/v$	$\in [1 \times 10^{-4}, 3 \times 10^1] \text{ m}^{-1}$
$\hat{E}_r = E_r/v = E_\rho/(av)$	$[0, 10^{-3}] \text{ V s m}^{-2}$
MONKES resolution (n_l, n_θ, n_ζ)	(180, 39, 99)
yancc resolution ($n_\alpha, n_\theta, n_\zeta$)	(201, 31, 81)

Table 3: Parameters and resolutions for the W7-X monoenergetic benchmark.

For a performance comparison, we re-implement the MONKES algorithm in JAX and compare time and memory required to solve the monoenergetic equation. Both codes run on the same hardware (Nvidia A100 GPU) to isolate algorithmic differences (comparing **yancc**’s finite difference discretization and Krylov/multigrid solver to the fully spectral discretization and block tridiagonal solver in MONKES). For this problem, **yancc** required 4 GB of memory, while MONKES required only 1.4 GB. This is likely due to the use of a Legendre discretization in MONKES, and the fact that the monoenergetic coefficients depend only on the first 2 Legendre modes, so very little storage is required. The tradeoff is that MONKES must invert a large number of dense matrices of size $(n_\theta n_\zeta) \times (n_\theta n_\zeta)$, which costs $O(n_\theta^3 n_\zeta^3)$, while **yancc** requires only operations that scale linearly with resolution. A timing comparison is shown in Figure 6, showing **yancc** is $2\times$ – $4\times$ faster for this problem over most of the collisionality range, though slowing down to roughly equal at the lowest collisionality. On problems with simpler geometry that require less resolution in the two flux surface angles, the MONKES algorithm will likely be faster. As for the discretization and resolution required, in this case they are broadly similar despite MONKES using a spectral discretization in all 3 coordinates. This is likely because the solutions are not sufficiently smooth for spectral convergence to show an advantage. If higher accuracy is desired then a fully spectral approach would be more efficient asymptotically, but at the level of accuracy common in optimization and analysis there is little difference.

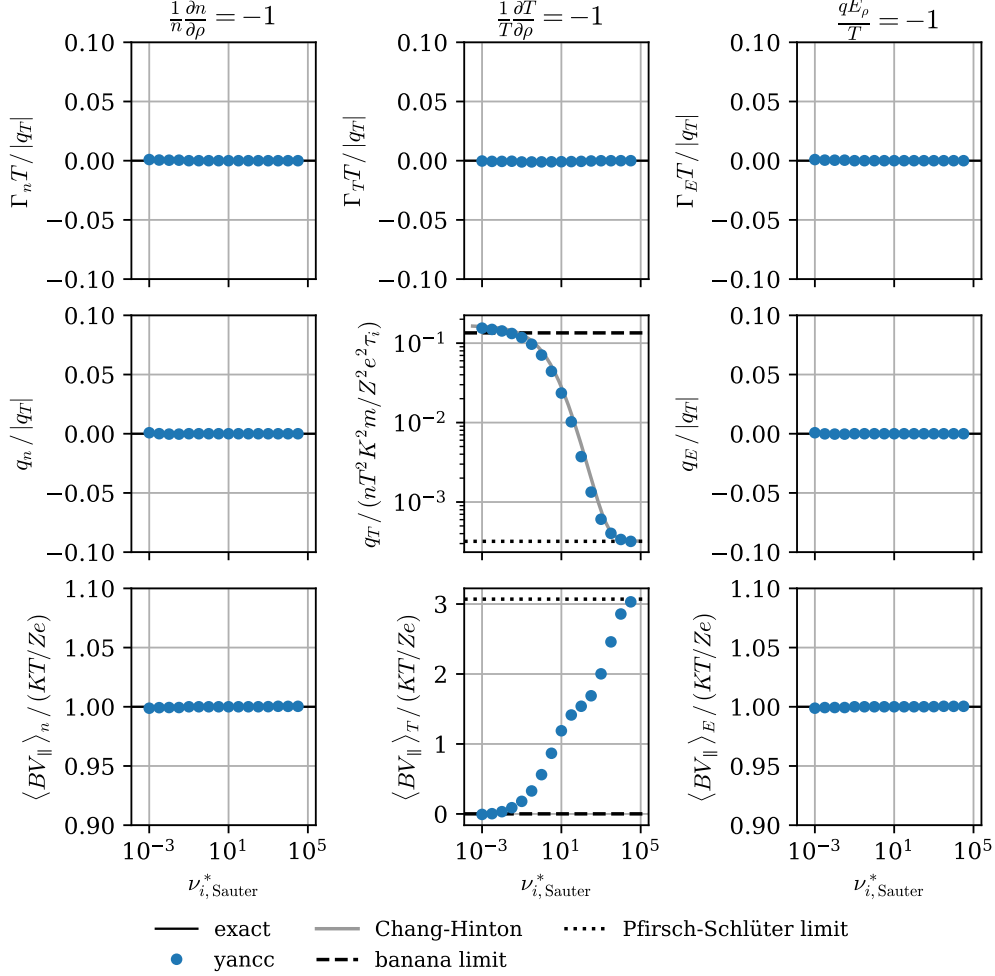


Figure 4: Ion transport matrix of a circular tokamak at $\epsilon = 0.01$ against the Sauter collisionality $\nu_{i,\text{Sauter}}^* = qR_0/(\epsilon^{3/2}v_T\tau_i)$, for which the banana regime is $\nu_{i,\text{Sauter}}^* \ll 1$ and the Pfirsch–Schlüter regime is $\nu_{i,\text{Sauter}}^* \gg \epsilon^{-3/2} = 10^3$. Columns are the three drives, each normalized to unit A_1 or $\partial_\rho \ln T$; rows are the radial particle flux, the radial conductive heat flux and the parallel flow. Flows are normalized to the diamagnetic flow $KT/(Ze)$ and the temperature-driven heat flux to $nT^2K^2m/(Z^2e^2\tau_i)$; quantities whose exact value vanishes are normalized to the temperature-driven flux of the same point. Solid lines are exact values, dashed the banana limit, dotted the Pfirsch–Schlüter limit, and gray the Chang–Hinton interpolation [13].

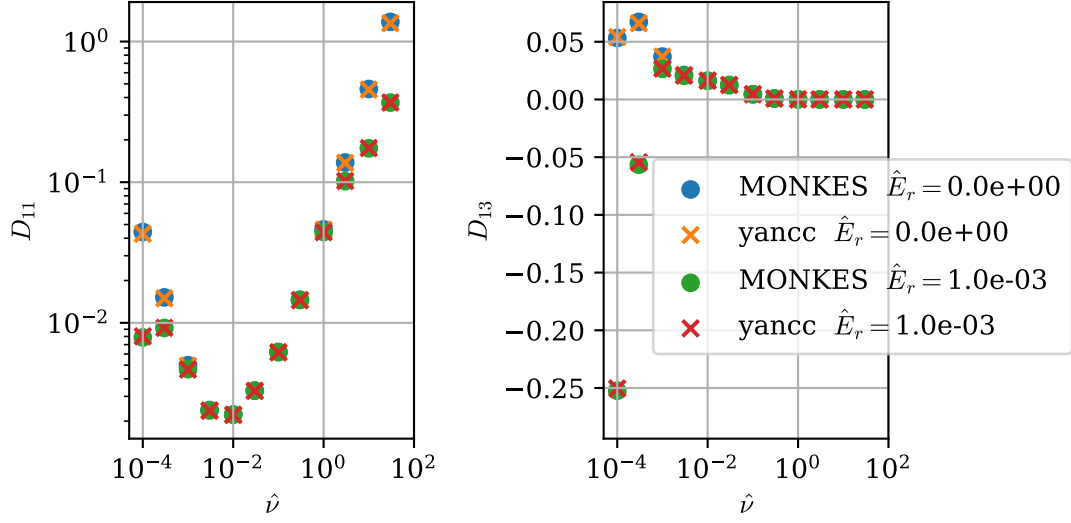


Figure 5: Monoenergetic transport coefficients D_{ij} for the W7-X high mirror configuration, comparing **yancc** against MONKES over a range of collisionality and electric field.

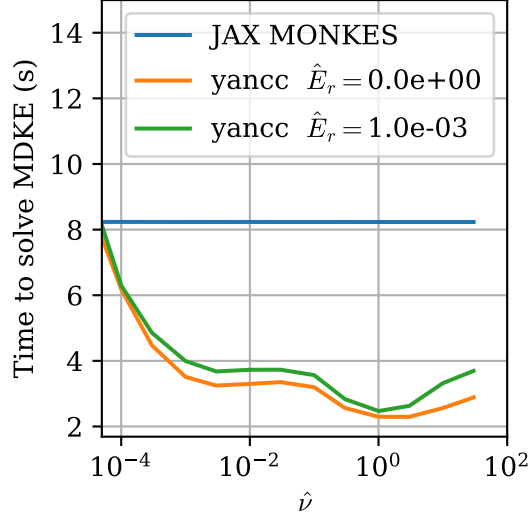


Figure 6: Timing comparison between **yancc** and a JAX re-implementation of the MONKES algorithm for the monoenergetic problem, as a function of collisionality.

5.3 Single-species density scan

For the full drift kinetic equation, we compare **yanc** to SFINCS [42], here for a single species scan in density / collisionality using an NCSX [73] like equilibrium. The parameters are summarized in Table 4, and a comparison of the output fluxes (particle flux $\langle \Gamma \rangle$, heat flux $\langle Q \rangle$, and parallel flow $\langle V_{\parallel} B \rangle$) is shown in Figure 7, where again both codes are converged to within 1%.

ρ	0.5
$E_r = E_{\rho}/a$	−1 kV/m
T	800 eV
n	$\in [1.5 \times 10^{19}, 1.5 \times 10^{22}] \text{ m}^{-3}$
$T^{-1} \partial_{\rho} T$	−0.81
$n^{-1} \partial_{\rho} n$	−0.86
$\nu^* = \frac{R_0 \nu_D}{v_{th\ell}}$	$5.2 \times 10^{-3} - 4.1 \times 10^0$
Species	Hydrogen
SFINCS resolution $(n_x, n_{\xi}, n_{\theta}, n_{\zeta})$	(7, 141, 25, 81)
yanc resolution $(n_x, n_{\alpha}, n_{\theta}, n_{\zeta})$	(7, 121, 43, 65)

Table 4: Parameters and resolutions for the single-species NCSX density scan.

A timing comparison is shown in Figure 8 across the full range of collisionality. For this case SFINCS was run on 128 CPU cores, while **yanc** was run on a single Nvidia A100 GPU, both on the Perlmutter cluster at NERSC and both using 64 bit floats. We see a speedup of roughly $5\times$ across the full range, up to nearly 2 orders of magnitude at higher collisionality where SFINCS slows down significantly. This is likely because the default preconditioner in SFINCS drops coupling in the speed coordinate, effectively ignoring energy scattering which becomes important at high collisionality. On the other hand **yanc** remains relatively flat, showing only modest slowdown at low or high collisionality. The times for **yanc** do not include the cost of just-in-time compilation in JAX (usually under 30 s) which must be paid once for a given resolution, after which the JIT compiled code is re-used across the scan. This is most representative of real world use where the DKE is solved repeatedly for parameter scans or optimization tasks.

Due to the different hardware this is not a strictly fair comparison, but to get a rough estimate we can compare peak theoretical flops. Assuming we could run SFINCS on GPU with the same overall efficiency as on CPU (unlikely due to the poor scaling of sparse matrix factorizations on GPU), we would expect a speedup of roughly 3 going from 128 CPU to $1\times$ A100. We can then see that the speedup from **yanc** is due to both being able to run on GPU as well as purely algorithmic improvements that are independent of hardware. In addition to the speed improvement, **yanc** also shows a clear win on memory usage. For these runs, SFINCS needed more than 50 GB, while **yanc** needed only 6 GB, which is especially important for GPU based codes since GPU memory is often a bottleneck.

5.4 Two-species E_{ρ} scan

To confirm the correctness for the 2 species case with full interspecies collisions and the ambipolarity condition we consider the same NCSX like equilibrium but perform a scan in radial electric field for 2 species, electrons and hydrogen. The parameters are given in Table 5 and the results shown in Figure 9. As before both codes are converged to within 1% for the particle and heat fluxes and parallel flows (particle flux and heat flux agree to $<1\%$, the parallel flow is often the hardest to converge). Composite quantities such as the radial and parallel currents are also shown. The parallel

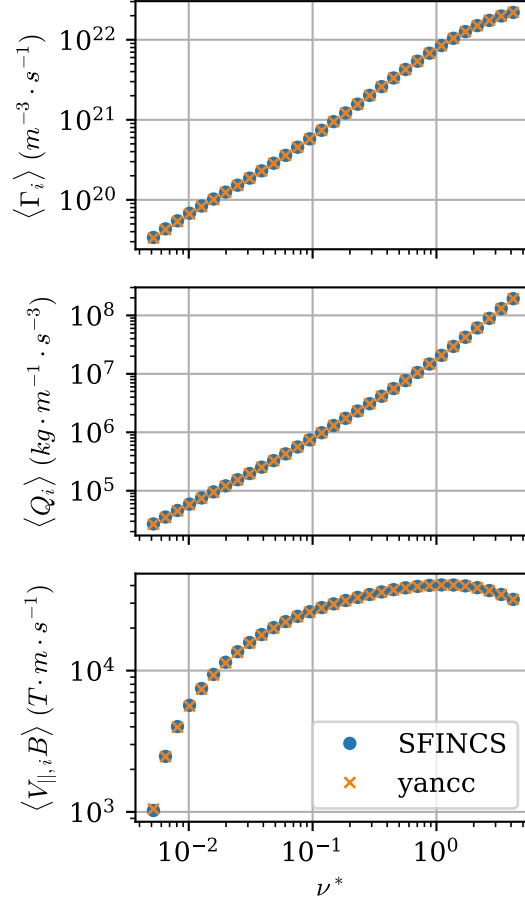


Figure 7: Output fluxes (particle flux $\langle \Gamma_i \rangle$, heat flux $\langle Q_i \rangle$, and parallel flow $\langle V_{\parallel i} B \rangle$) for the single-species NCSX density scan, comparing `yancc` and SFINCS.

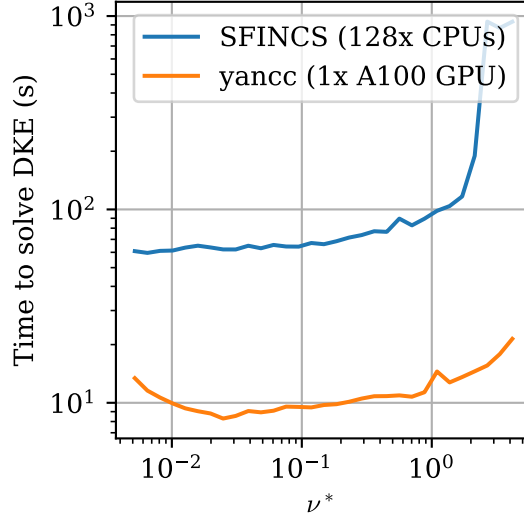


Figure 8: Timing comparison between **yamcc** (1× A100 GPU) and SFINCS (128 CPU cores) for the single-species NCSX density scan, across collisionality.

current shows slightly larger disagreement ($\sim 5\%$), which is to be expected as it is the difference of parallel flows.

As with the comparison to MONKES, we see that **yamcc** requires broadly similar resolution to SFINCS, though in some cases SFINCS can use slightly coarser resolution in the flux surface coordinates. This is likely due to the centered differences that SFINCS uses, which avoids the extra numerical diffusion caused by upwinding. The cost of this can be stability, and we find that in some cases **yamcc** can get away with coarser resolution.

ρ	0.5
$E_r = E_\rho/a$	$\in [-8, 8]$ kV/m
$T_H = T_e$	800 eV
$n_H = n_e$	$1.5 \times 10^{20} \text{ m}^{-3}$
$T_H^{-1} \partial_\rho T_H = T_e^{-1} \partial_\rho T_e$	-0.81
$n_H^{-1} \partial_\rho n_H = n_e^{-1} \partial_\rho n_e$	-0.86
$\nu_H^* = \frac{R_0 \nu_{D,H}}{v_{th,H} \ell}$	5.0×10^{-2}
$\nu_e^* = \frac{R_0 \nu_{D,e}}{v_{th,e} \ell}$	1.3×10^{-1}
Species	Hydrogen, Electrons
SFINCS resolution ($n_x, n_\xi, n_\theta, n_\zeta$)	(7, 61, 15, 31)
yamcc resolution ($n_x, n_\alpha, n_\theta, n_\zeta$)	(7, 61, 25, 37)

Table 5: Parameters and resolutions for the two-species NCSX radial electric field scan.

5.5 Ambipolar rootfinding in W7-X

As a test of the ability to find the ambipolar electric field, we consider again the high mirror configuration of W7-X, and take a surface close to the axis with $T_e \gg T_i$ and zero density gradient

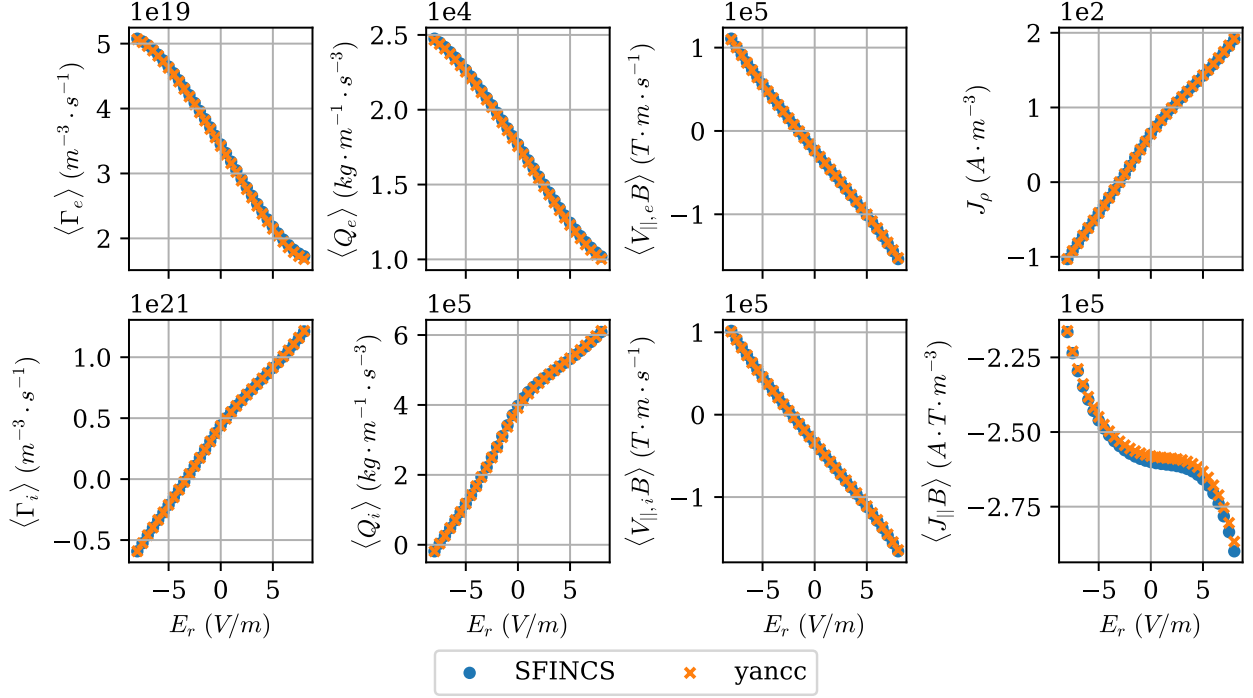


Figure 9: Two-species (hydrogen and electron) radial electric field scan for the NCSX like equilibrium, comparing **yancc** and SFINCS on particle flux (Γ_e , Γ_i), heat flux (Q_e , Q_i), parallel flow ($V_{||,e}B$, $V_{||,i}B$), radial current (J_ρ) and bootstrap current ($J_{||}B$).

(full parameters given in Table 6), where we expect to find multiple roots. We see that **yancc** is able to find all three roots as shown in Figure 10, corresponding to the ion branch, the electron branch, and the unstable branch in between. For this problem we gave no initial guess, only very wide bounds, corresponding to $E^* = \pm 0.1$, where E^* is the electric field normalized to its resonant value (as $E^* \rightarrow \pm 1$ the drift ordering assumed by the DKE breaks down and nonphysical roots can appear). This should contrast with existing workflows for finding multiple roots which generally involve a dense scan in E_ρ , followed by visual inspection to look for likely root locations and then local root finding from those initial points or brackets. For this search, **yancc** required 42 solutions of the DKE to find all three roots, which is already comparable to the number required for a dense scan to first bracket the roots in a standard workflow, and has the advantage of being fully automated.

6 Conclusion

We have presented **yancc**, a new solver for the drift kinetic equation designed for GPU acceleration and automatic differentiability. The code solves both the monoenergetic and full four-dimensional forms of the equation, using a mixed pseudo-spectral/finite difference discretization with a multigrid-preconditioned Krylov solver. The numerical benchmarks demonstrate agreement with MONKES and SFINCS to within 1% across a wide range of collisionalities, electric fields, and magnetic geometries, for both single-species and multi-species plasmas, and for both stellarator and tokamak configurations.

In terms of performance, **yancc** achieves roughly an order of magnitude speedup over SFINCS on a per-scan basis while using an order of magnitude less memory, with the gap widening at high

ρ	0.1919
E_ρ	$\in [-5.7 \times 10^4, 5.7 \times 10^4]$ V
T_H	1150 eV
T_e	2650 eV
$n_H = n_e$	$6 \times 10^{19} \text{ m}^{-3}$
$T_H^{-1} \partial_\rho T_H$	-0.869
$T_e^{-1} \partial_\rho T_e$	-1.73
$n_H^{-1} \partial_\rho n_H = n_e^{-1} \partial_\rho n_e$	0.0
$\nu_H^* = \frac{R_0 \nu_{D,H}}{v_{\text{th},H} \iota}$	2.1×10^{-2}
$\nu_e^* = \frac{R_0 \nu_{D,e}}{v_{\text{th},e} \iota}$	9.4×10^{-3}
Species	Hydrogen, Electrons
yanc c resolution ($n_x, n_\alpha, n_\theta, n_\zeta$)	(6, 181, 97, 31)

Table 6: Parameters and resolutions for the W7-X ambipolar electric field solve.

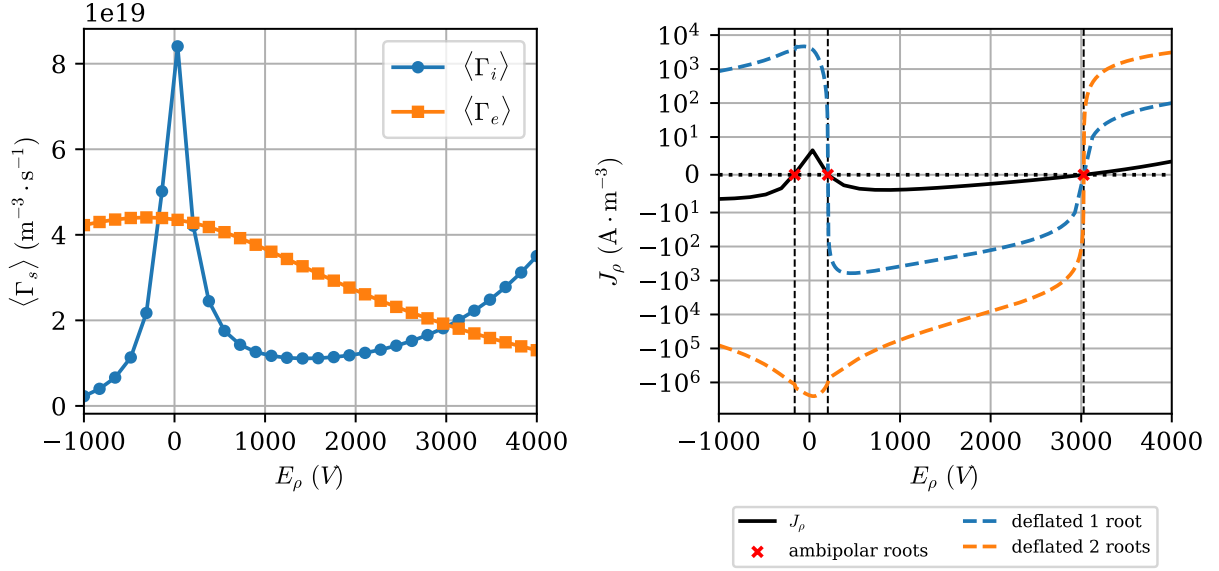


Figure 10: Left: Particle fluxes (electron and ion) as a function of E_ρ for the W7-X high mirror configuration. Right: The radial current J_ρ and the location of the found ambipolar roots. Also shown are the modified curves after deflating each found root. The bounds used for the rootfinding were more than $10\times$ wider than the region shown, but **yanc**c is able to efficiently explore this space and find the desired roots.

collisionality where the SFINCS preconditioner degrades. The runtime remains nearly flat across a wide range of collisionality, unlike existing solvers whose iteration counts grow significantly at extremes of collisionality. This robustness makes `yancc` practical for production scans covering the full parameter space of a given device.

The speed and differentiability of `yancc` open up several applications beyond standalone neoclassical calculations. In stellarator optimization, neoclassical transport coefficients must be evaluated many times as the magnetic geometry is varied, and gradients of these coefficients with respect to geometry parameters enable efficient gradient-based optimization over the full configuration space. The low memory footprint and GPU acceleration support the high throughput needed for uncertainty quantification through Monte Carlo sampling over input parameter distributions. For integrated modeling and profile prediction, `yancc` is fast enough to compute neoclassical coefficients on-the-fly within a transport solver, replacing the precomputed fit functions or interpolated databases that are commonly used today. The differentiability also enables adjoint-based sensitivity analysis, where the derivative of a quantity of interest with respect to all input parameters can be computed in a single solve.

Several directions for future development are planned. Adaptive mesh refinement in pitch angle and real space is being explored to efficiently resolve the trapped-passing boundary layers, which can span several orders of magnitude in width depending on collisionality, without requiring uniformly fine grids across the full angular domain. Incorporating corrections for strong radial electric fields and flows near quasisymmetry, building on the work of [53], will extend the validity of the code to regimes where the conventional ρ_* ordering assumptions begin to break down. When considering impurities, additional terms in the DKE can become important, such as tangential drifts or the in-surface variation of the electrostatic potential [23], and including these terms is planned. Together, these developments will make `yancc` a versatile tool for neoclassical transport calculations across the full range of fusion-relevant regimes, from conceptual design through discharge analysis.

Data Availability

The source code and data supporting this work are available at <https://github.com/f0uriest/yancc>.

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A Rosenbluth potential implementation

The Rosenbluth potentials satisfy Poisson type equations:

$$\nabla_v^2 H_s = -4\pi f_{1,s} \quad (59)$$

$$\nabla_v^2 G_s = 2H_s \quad (60)$$

We can expand these in Legendre polynomials in $\xi = -\cos \alpha$:

$$H_s(\alpha, x_s) = \sum_l P_l(-\cos \alpha) H_{s,l}(x_s) \quad (61)$$

$$G_s(\alpha, x_s) = \sum_l P_l(-\cos \alpha) G_{s,l}(x_s) \quad (62)$$

And similarly for $f_{1,s}$:

$$f_{1,s}(\alpha, x_s) = \sum_l P_l(-\cos \alpha) f_{1,s,l}(x_s) \quad (63)$$

The Legendre harmonics of the potentials are then given by the Green's function:

$$H_{s,l}(x_s) = \frac{4\pi v_{\text{th},s}^2}{2l+1} \left[\frac{1}{x_s^{l+1}} I_{2,l}(x_s) + x_s^l I_{1,l}(x_s) \right] \quad (64)$$

$$G_{s,l}(x_s) = -\frac{4\pi v_{\text{th},s}^4}{4l^2-1} \left[x_s^l I_{3,l}(x_s) - \frac{2l-1}{2l+3} x_s^{l+2} I_{1,l}(x_s) - \frac{2l-1}{2l+3} \frac{1}{x_s^{l+1}} I_{4,l}(x_s) + \frac{1}{x_s^{l-1}} I_{2,l}(x_s) \right] \quad (65)$$

with

$$I_{1,l}(x_s) = \int_{x_s}^{\infty} z^{-l+1} f_{1,s,l}(z) dz \quad (66)$$

$$I_{2,l}(x_s) = \int_0^{x_s} z^{l+2} f_{1,s,l}(z) dz \quad (67)$$

$$I_{3,l}(x_s) = \int_{x_s}^{\infty} z^{-l+3} f_{1,s,l}(z) dz \quad (68)$$

$$I_{4,l}(x_s) = \int_0^{x_s} z^{l+4} f_{1,s,l}(z) dz \quad (69)$$

We can then express $f_{1,s,l}$ using weighted Maxwell polynomials $L_k(x_s)$:

$$f_{1,s,l}(x_s) = \sum_k f_{1,s,l,k} L_k(x_s) \exp(-x_s^2) \quad (70)$$

and define

$$I_{1,l,k}(x_s) = \int_{x_s}^{\infty} z^{-l+1} L_k(z) \exp(-z^2) dz \quad (71)$$

$$I_{2,l,k}(x_s) = \int_0^{x_s} z^{l+2} L_k(z) \exp(-z^2) dz \quad (72)$$

$$I_{3,l,k}(x_s) = \int_{x_s}^{\infty} z^{-l+3} L_k(z) \exp(-z^2) dz \quad (73)$$

$$I_{4,l,k}(x_s) = \int_0^{x_s} z^{l+4} L_k(z) \exp(-z^2) dz \quad (74)$$

so that

$$I_{1,l}(x_s) = \sum_k f_{1,s,l,k} I_{1,l,k}(x_s) \quad (75)$$

$$I_{2,l}(x_s) = \sum_k f_{1,s,l,k} I_{2,l,k}(x_s) \quad (76)$$

$$I_{3,l}(x_s) = \sum_k f_{1,s,l,k} I_{3,l,k}(x_s) \quad (77)$$

$$I_{4,l}(x_s) = \sum_k f_{1,s,l,k} I_{4,l,k}(x_s) \quad (78)$$

We can then define

$$H_{s,l,k}(x_s) = \frac{4\pi}{2l+1} \left[\frac{1}{x_s^{l+1}} I_{2,l,k}(x_s) + x_s^l I_{1,l,k}(x_s) \right] \quad (79)$$

$$G_{s,l,k}(x_s) = -\frac{4\pi}{4l^2-1} \left[x_s^l I_{3,l,k}(x_s) - \frac{2l-1}{2l+3} x_s^{l+2} I_{1,l,k}(x_s) - \frac{2l-1}{2l+3} \frac{1}{x_s^{l+1}} I_{4,l,k}(x_s) + \frac{1}{x_s^{l-1}} I_{2,l,k}(x_s) \right] \quad (80)$$

To compute the integrals $I_{i,l,k}$, we use a fixed composite Gauss–Legendre quadrature rule of 32 nodes per panel. For the integrals over $[0, x_s]$ (I_2 and I_4), we use two uniform panels on $[0, \min(x_s, 10)]$, beyond which the integrand is negligible. The integrals over $[x_s, \infty)$ (I_1 and I_3) contain negative powers of z that are sharply peaked near $z = x_s$ when $x_s \ll 1$, so we use four geometrically graded panels on $[x_s, \max(x_s, 1)]$, followed by two uniform panels covering the Maxwellian tail up to $\max(x_s, 1) + \min(8, 22/x_s + 1)$. Compared to high precision reference values, this rule attains an error within a small multiple of the best achievable in double precision (machine precision times the integral of the absolute value of the integrand) for n_x up to 20 and $l_{\max}$ up to 16 which covers all but the most extreme cases. Beyond this we use an adaptive quadrature routine from quadax [16].

The distribution function f_s is stored on a grid in (x, α) as $f_{s,x,\alpha}$, and the Green's functions $H_{s,l,k}$ and $G_{s,l,k}$ are stored on the same speed grid, for each cross species interaction as $H_{s,s',x',l,k}$ and $G_{s,s',x',l,k}$ (where $x' = x_s v_{\text{th},s}/v_{\text{th},s'}$, the potential for species s evaluated on the speed grid for species s'). The Rosenbluth potentials can then be evaluated as simple tensor contractions:

$$H_{s,s',x',\alpha'} = \sum_{l,k,x,\alpha} T_{l,\alpha'}^{-1} H_{s,s',x',l,k} V_{k,x} T_{l,\alpha} f_{s,x,\alpha} \quad (81)$$

$$G_{s,s',x',\alpha'} = \sum_{l,k,x,\alpha} T_{l,\alpha'}^{-1} G_{s,s',x',l,k} V_{k,x} T_{l,\alpha} f_{s,x,\alpha} \quad (82)$$

where $T_{l,\alpha}$ is the change of basis matrix transforming between pitch angle α and Legendre index l and $V_{k,x}$ is the change of basis matrix between speed x and Maxwell polynomial index k . Derivatives of the potentials with respect to speed required for the collision operator are obtained analytically and precomputed/applied in a similar fashion. The cost of these tensor operations is quite low, as the resolution in x is fairly coarse ($n_x \sim 5$ – 10), and generally only a small number of Legendre modes are required to represent the potentials (generally $l_{\max} \sim 4$ – 6).

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